CPGE PTSI-PT Lycée Jean Zay - Thiers

TECHNOLOGIE

LIAISON ENCASTREMENT

Mémo

Compétences visées: F2-01, F2-03 v1.4

Lycée Jean Zay – 21 rue Jean Zay – 63300 Thiers – Académie de Clermont-Ferrand

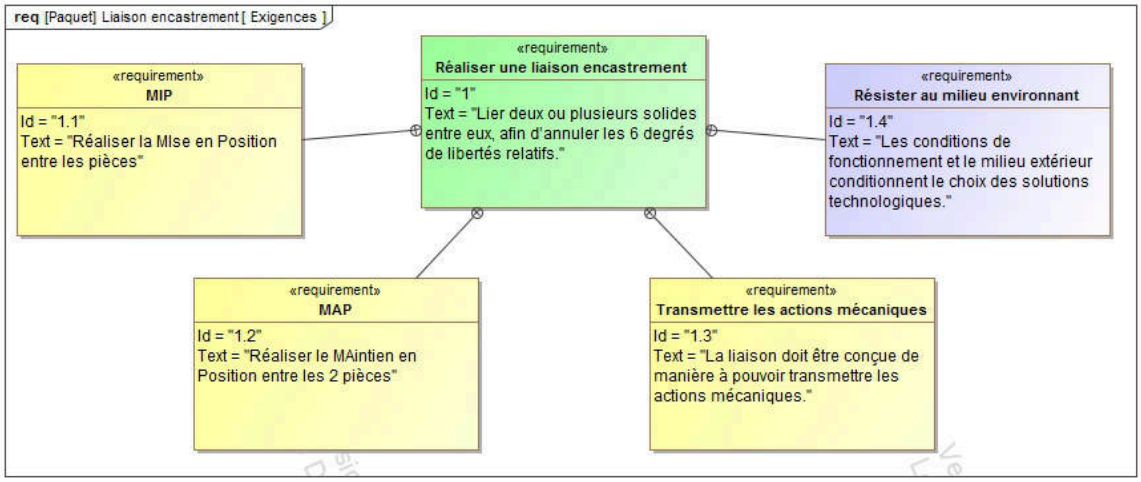
1 Introduction

Du point de vue du modèle cinématique, une liaison encastrement n'autorise aucun degré de liberté entre deux pièces constitutives d'un mécanisme. Le langage courant (mais inadapté) utilise « liaison complète ».

Une liaison complète étant une liaison encastrement, son torseur cinématique relatif est nul. Nous savons d'autre part que le torseur des actions mécaniques transmissibles par cette liaison est complet et de la forme :

$$\{\mathcal{T}_{2\to1}\} = \begin{Bmatrix} X_{21} & L_{21} \\ Y_{21} & M_{21} \\ Z_{21} & N_{21} \end{Bmatrix}_{\substack{A \\ b}}$$

2 Exigences relatives à la liaison encastrement

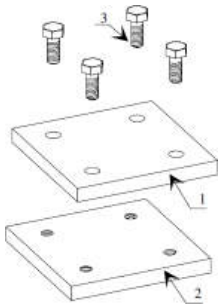
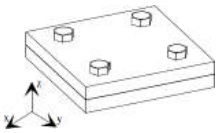
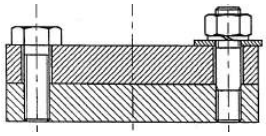
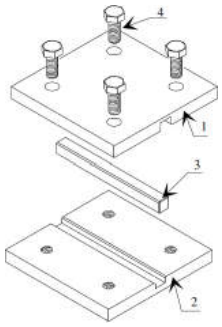
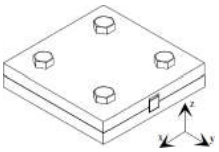
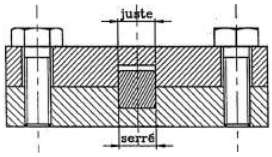
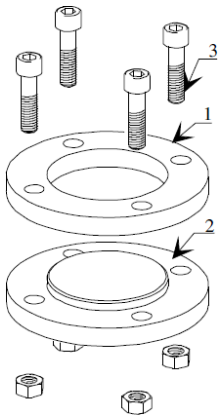
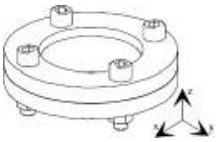
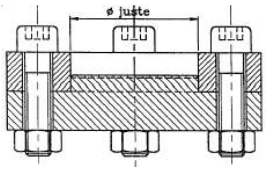
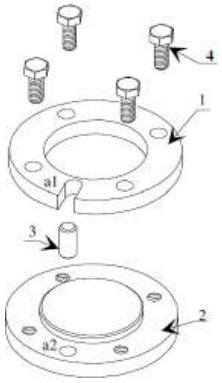
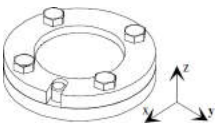
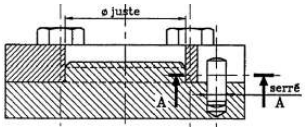


Attention

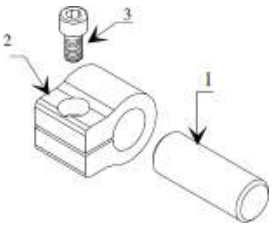
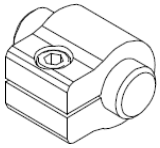
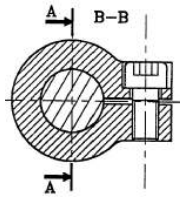
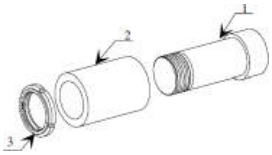
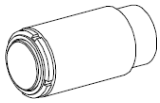
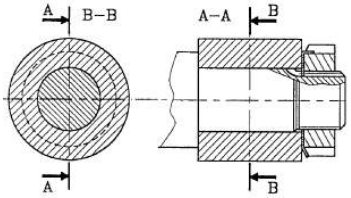
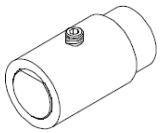
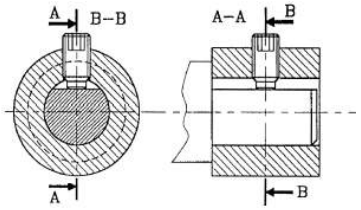
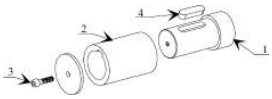
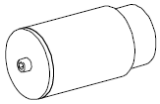
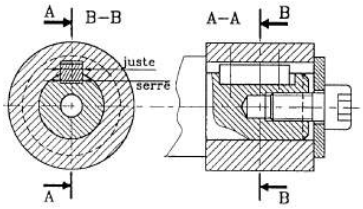
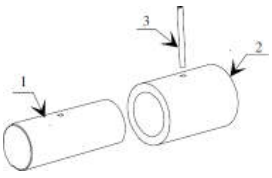
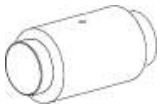
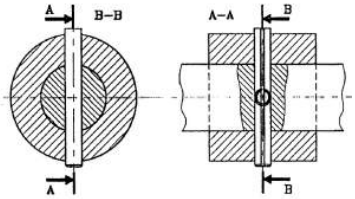
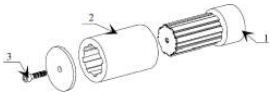
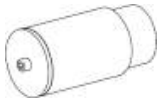
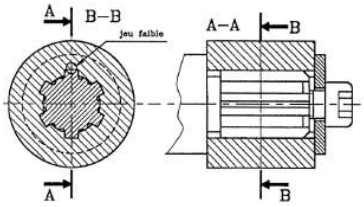
Ce mémo se bornera à l'étude des solutions technologues réalisant des liaisons encastrement **démontables**.

3 Liaisons démontables

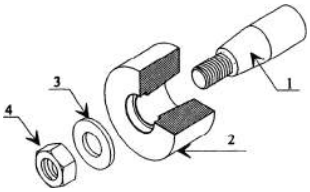
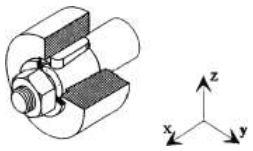
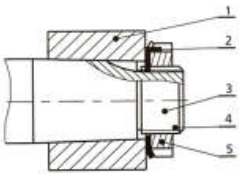
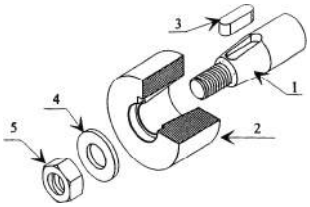
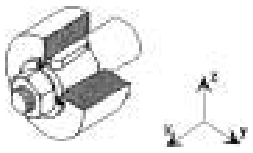
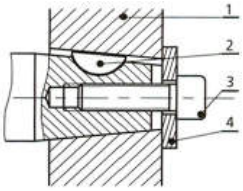
3.1 À partir de surfaces planes prépondérantes

Vue éclatée	Vue 3D	Vue en projection	Analyse
			MIP : Appui plan MAP : Vis Les ddl restant (T_x, T_y, R_z) sont supprimés par adhérence.
			MIP : 2 appuis plans orthogonaux MAP : Vis Le ddl restant (T_y) est supprimé par adhérence.
			MIP : Appui plan + centrage court MAP : Vis/écrous Le ddl restant (R_z) est supprimé par adhérence.
			MIP : Appui plan + centrage court + locating (isostatique) MAP : Vis

3.2 À partir de surfaces cylindriques prépondérantes

Vue éclatée	Vue 3D	Vue en projection	Analyse
			MIP : Centrage long MAP : Pincement (vis) Les ddl restant (T_x, R_x) sont supprimés par adhérence.
			MIP : Appui plan + centrage long MAP : Écrou à encoches (SKF) Le ddl restant (R_x) est supprimé par adhérence.
			MIP : Appui plan + centrage long + vis de pression (isostatique) MAP : Vis
			MIP : Appui plan + centrage long + clavette (isostatique) MAP : Vis
			MIP : Appui plan + centrage long + goupille (isostatique) MAP : Réalisé par la MIP
			MIP : Appui plan + cannelures ou dentelures (isostatique) MAP : Vis

3.3 À partir de surfaces coniques prépondérantes

Vue éclatée	Vue 3D	Vue en projection	Analyse
			MIP : Emmanchement conique MAP : Écrou à encoches (SFK) Le ddl restant (R_x) est supprimé par adhérence.
			MIP : Emmanchement conique + clavette (isostatique) MAP : Vis

3.4 Bilan des dispositions constructives envisageables

- **MIP :** emboitement cylindrique, emboitement conique, emboitement rectangulaire, goupilles, entretoises, pions de positionnement,...
- **MAP :** éléments filetés (vis, écrou, boulon, goujon, vis pointeau...), anneaux élastiques (circlips), tampons tangeants, pincement, coincement conique, ajustement serré...
- **Transmission d'actions mécanique :** goupilles, clavettes, cannelures, dentelures, vis/pointeau. (ces éléments peuvent participer à la MIP). Dans le cas d'efforts peu importants, l'adhérence créée par le MAP peut-être suffisante.

4 Liaisons permanentes (non démontables)

Les solutions n'offrant pas de possibilité de démontage (sans destruction ou détérioration des organes) sont, entre autres :

- le soudage (ou brasage),
- le collage,
- le rivetage,
- le frettage (montage en force en refroidissant l'arbre ou en chauffant l'alésage),
- le clinchage (assemblage de 2 tôles par emboutissage).

5 Analyse de quelques solutions pour l'encastrement de pièces cylindriques

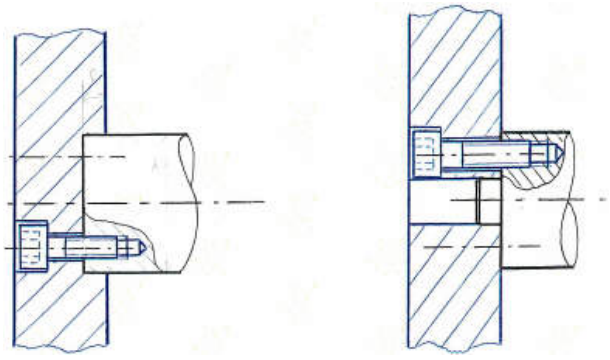
Un encastrement se définit en premier par ses surfaces de mise en position qui seront ajustées ⌀H h.

5.1 Appui plan + L.A (appui plan + centrage court)

On se considère dans ce cas quand :

$$\frac{L}{D} \leq 0,5$$

À privilégier pour les pièces $\varnothing > 20$. Il est sans jeu, adaptable aux efforts à transmettre (nombre et diamètre des vis) et les surfaces fonctionnelles sont usinables en une seule phase.

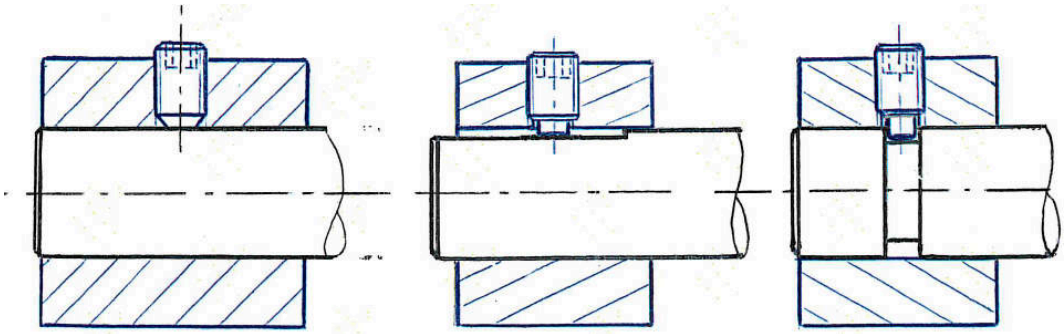


5.2 Pivot glissant + arrêt axial + arrêt en rotation (centrage long)

On se considère dans ce cas quand : $\frac{L}{D} \geq 1$

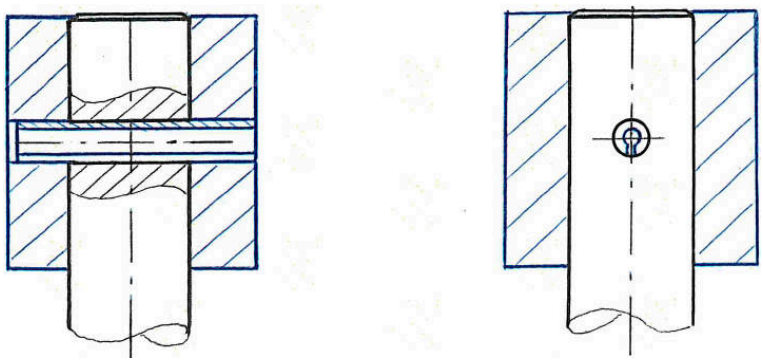
5.2.1 Vis de pression

Effort transmis faible, peu fiable, mais les positions axiale et angulaire sont réglables.



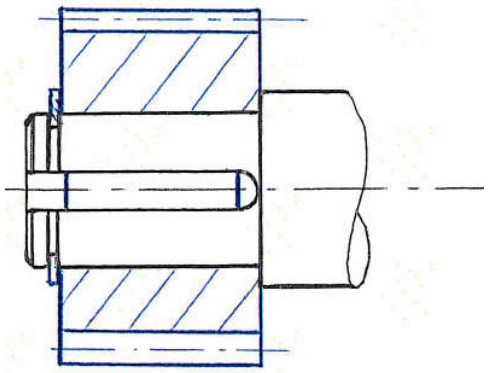
5.2.2 Goupille élastique

Effort transmis réduit mais calculable et fiable. Liaison sans jeu.

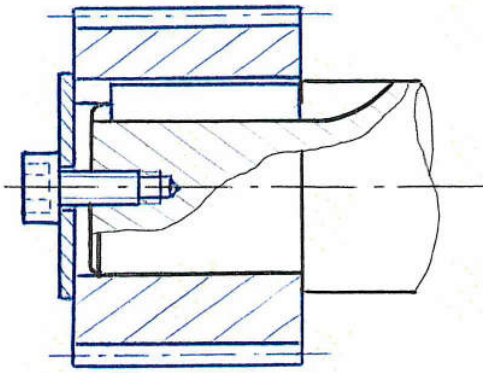


5.2.3 Clavette

Nécessite un épaulement fixe et un arrêt démontable. Liaison avec jeu angulaire qui nécessite un brochage séparé dans l'alésage débouchant. Couple transmis moyen (en fonction de la longueur de la clavette).



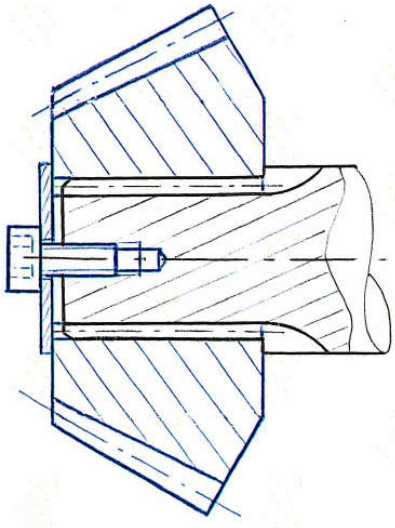
Arrêt par circlips : jeu axial



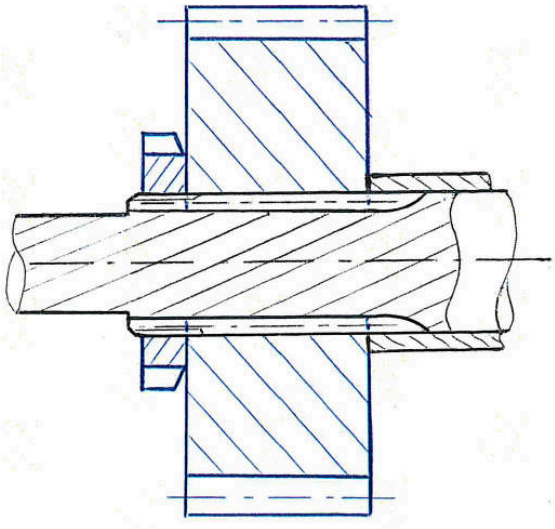
Arrêt par vis en bout : sans jeu axial

5.2.4 Cannelures

Nécessite un épaulement fixe et un arrêt démontable. Liaison avec jeu angulaire réduit qui nécessite un brochage séparé dans l'alésage débouchant. Couple transmis élevé en fonction de la longueur des cannelures.



Arrêt par vis en bout



Arrêt par écrou à encoche si en milieu d'arbre

Remarque

L'épaulement fixe est **toujours** à privilégier. Si le montage le nécessite, on peut le remplacer par une entretoise (attention cependant à bien arrêter l'entretoise en translation !).

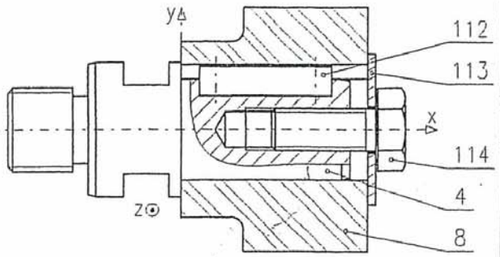


Figure 1

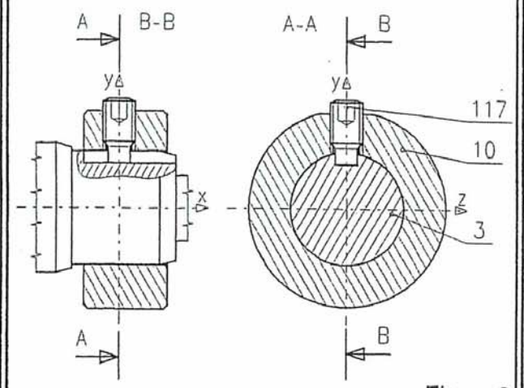


Figure 2

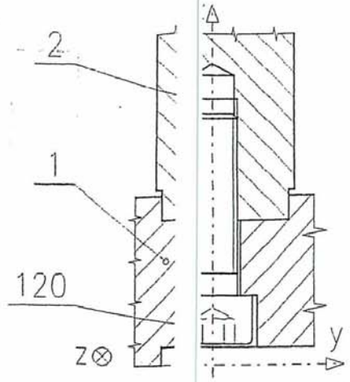


Figure 3

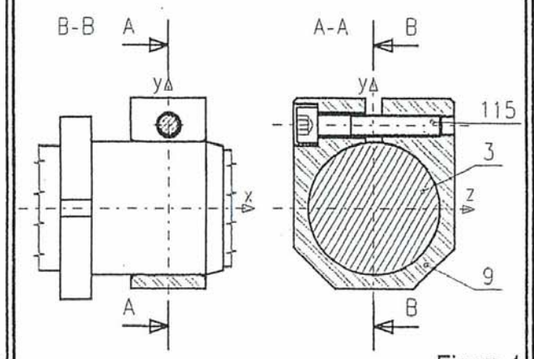


Figure 4

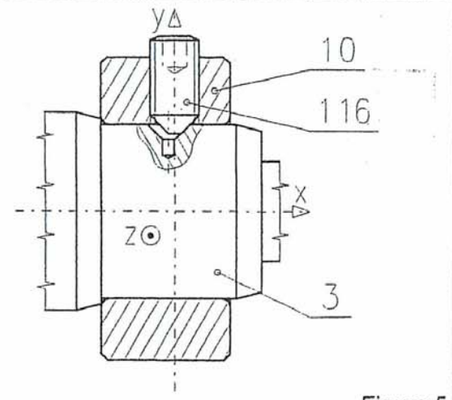


Figure 5

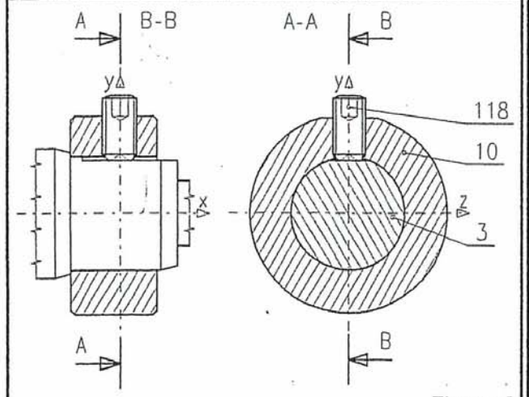


Figure 6

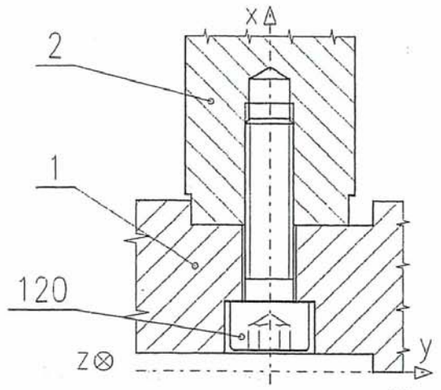
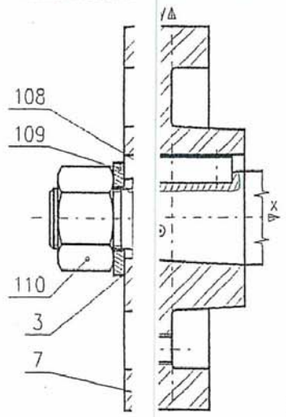


Figure 7



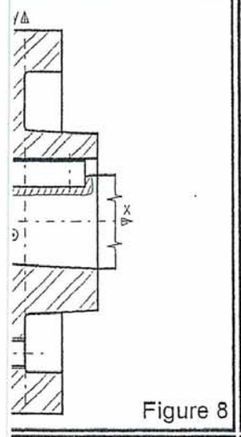


Figure 8

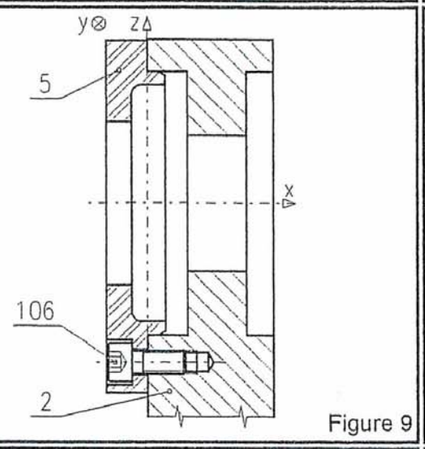


Figure 9

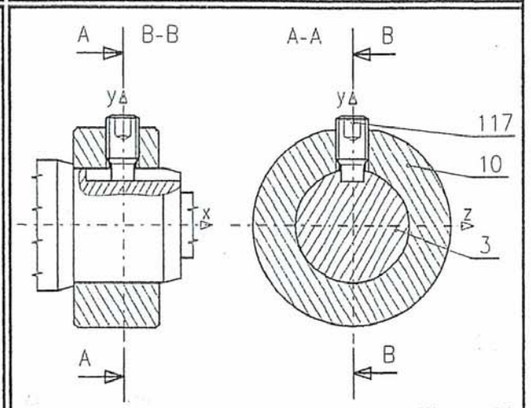


Figure 10

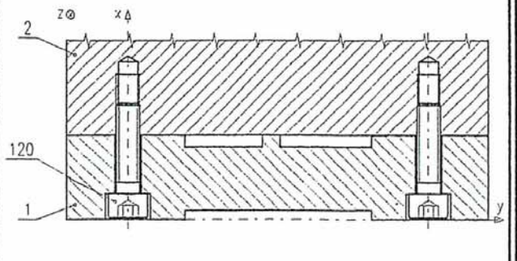


Figure 11

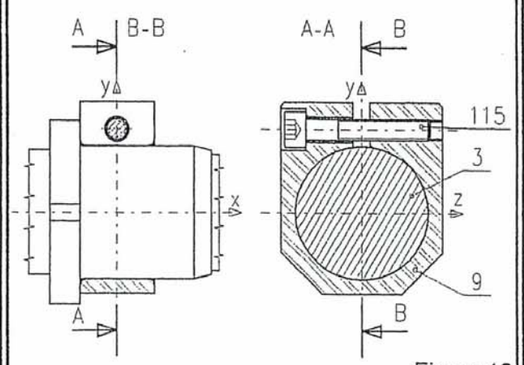
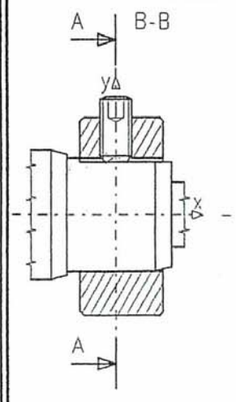


Figure 12



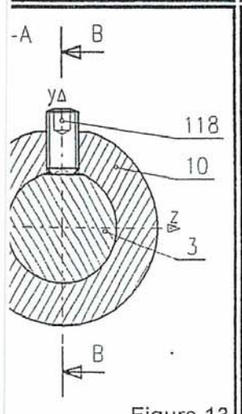


Figure 13

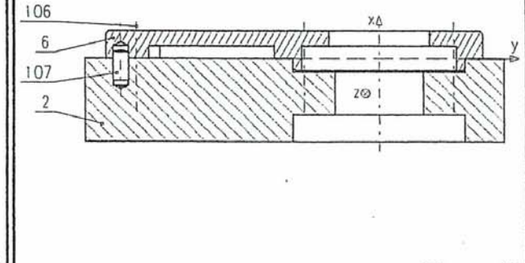


Figure 14

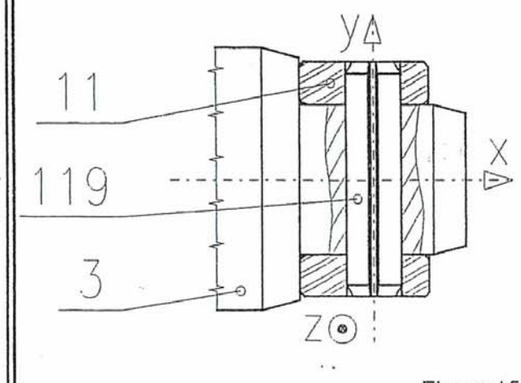


Figure 15

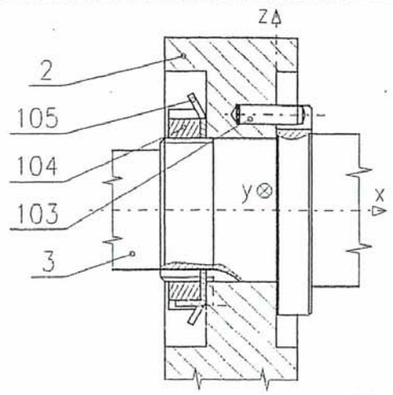


Figure 16

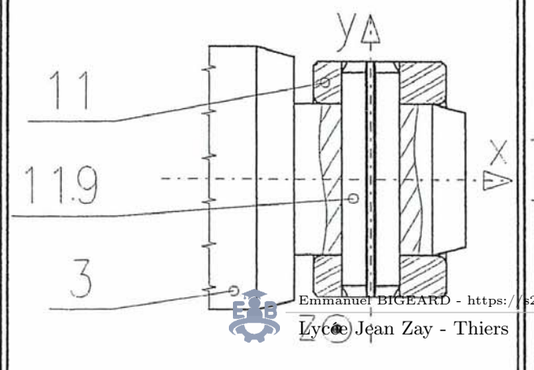
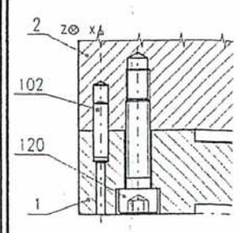


Figure 17



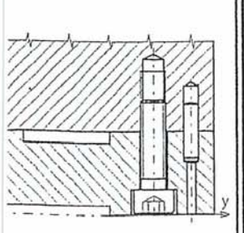


Figure 18

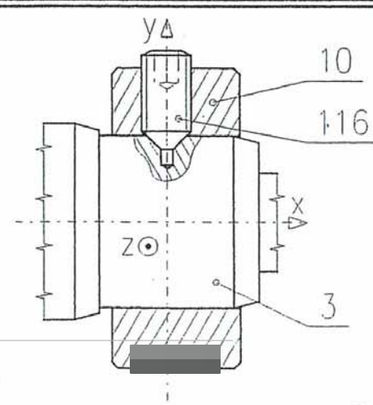


Figure 19

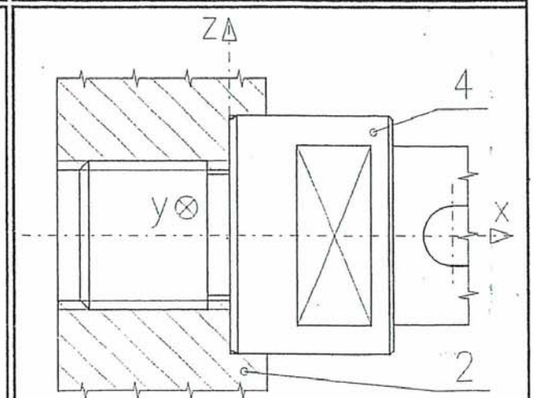


Figure 20

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